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Parent and carer perspectives on behavioural sleep management strategies for children with ADHD

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ABSTRACT

Objective/background: Attention-Deficit/Hyperactivity Disorder (ADHD) is common in children. Many children with ADHD have chronic insomnia, which is associated with negative outcomes for the child and family and can worsen ADHD symptoms. Behavioural sleep interventions can be effective for children with ADHD and are the recommended first-line approach to management. However, many parents/carers may not have access to, or struggle to implement, interventions in practice.

This study explored parents/carers views and experiences of: 1) behavioural sleep management strategies, 2) help-seeking for managing sleep problems and 3) perceptions about a potential digital sleep guide for managing sleep problems.

Patients/methods: A qualitative study using semi-structured interviews of parents/carers of children with ADHD/ADHD traits and chronic insomnia symptoms (CIS). Participants were recruited via social media and purposively sampled for diverse characteristics. Interviews were conducted by telephone/videocall, recorded and transcribed verbatim. Data were analysed using inductive thematic analysis.

Results: Eighteen parents/carers took part; two themes and six subthemes were developed. Despite valuing behavioural sleep strategies, parents/carers noted many challenges to engaging with them, including perceiving these strategies to be both ineffective for children with ADHD and not easy to implement in the real world. They described that tiredness, exhaustion, ADHD symptoms and challenging behaviour make parent engagement with strategies difficult. Parents/carers welcomed digital support in principle, particularly because they perceived health professional support to be lacking. However, they noted that digital interventions must address their key support needs regarding flexibility, realistic expectations and a pragmatic approach, as well as comradeship, support and testimonials from other parents/carers.

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Conclusion: Tailored but flexible support and reassurance are necessary for parents/carers of children with ADHD and CIS. Digital support must validate parent/carer experiences and include experiences from other parents/carers.

1. Introduction

1.1. Background

Attention-Deficit/Hyperactivity Disorder (ADHD) affects around 5% of school-aged children worldwide [1]. Characterised by developmentally inappropriate inattention, hyperactivity/impulsivity or both [2], ADHD can have a major impact on children and families [3]. Around 70% of parents of children with ADHD report that their child experiences sleep problems and individuals with ADHD have an eight-fold increased risk of a sleep disorder diagnosis (most commonly chronic insomnia), compared to those without ADHD [4].

Chronic insomnia (CI) disorder is defined by the International Classification of Sleep Disorders III [5] as difficulty with initiating/maintaining sleep or early morning waking, for at least three nights a week and for at least three months. Sleep problems in children can affect cognitive function, emotional regulation and increase risk of obesity [6,7]. They can also worsen ADHD symptoms [8] and impact parental sleep and quality of life [9,10].

Currently, many types of behavioural sleep interventions for CI in children are available [11,12]. These often include parent education, appropriate and consistent bedtime routines and appropriate sleep habits (e.g., improved physical and emotional sleep environment, reduced screen time and reduced caffeine in the evening). More specific sleep interventions include faded bedtime (delaying bedtime in the short term to gradually bring it forward over time), extinction-based strategies (such as camping-out; gradually increasing physical distance between the parent and child at bedtime, to promote independent sleeping) and positive reinforcement (rewards for positive sleep behaviours). Behavioural sleep interventions are effective for children with ADHD [13] - yet with small to moderate effect sizes and low certainty of evidence - and recommended as the first approach to CI [14].

Parenting interventions for sleep problems in children with ADHD can be effective at improving parent reported ADHD symptoms [15,16], emotional and physical wellbeing and relationships [17] when delivered by practitioners.

Whilst the literature suggests that various behavioural sleep interventions are effective, a qualitative study of 12 parents/carers of children with ADHD, conducted by French et al. [18] aimed to explore experiences of sleep difficulties. The findings highlighted parents usually try many types of strategies to improve their child's sleep, using a trial-and-error approach. Rowsell et al. [19], in a qualitative study conducted by the current research team, further found that repeated unsuccessful attempts to address children's sleep issues led to emotional distress and eventual normalisation of the problems. It is therefore important to explore in depth the barriers and facilitators to implementing specific approaches, to better support parents/carers to persist with appropriate strategies in the future. To date, there are no published studies from the UK exploring this topic.

Additionally, many services for children with ADHD, including those provided by the UK national health service (NHS), are overstretched and thus unable to provide timely support [20]. Furthermore, behavioural sleep interventions are resource intensive, and medication is often prescribed without due adherence to international guidance recommending first-line non-pharmacological interventions [14,21]. It is therefore important to also explore support parents need for their child's sleep problems, to guide the format and delivery of behavioural sleep interventions.

Digital interventions specifically designed to address sleep problems in children with ADHD may be low cost, with wide reach and represent an effective solution to current challenges in service provision. Indeed, a digital sleep intervention encompassing CBT for insomnia, aimed at a general population of children with insomnia and their parents, reduced sleep problems and was well accepted by parents [22]. Digital parenting interventions for ADHD, not specific to sleep problems, have also been suggested as acceptable to parents and to increase parenting efficacy and child social performance [23].

We therefore conducted the present study to inform the development of a digital sleep intervention for parents/carers of children with ADHD [24] (https://www.discasleep.org.uk/).

1.2. Study aims and research questions

To explore:1) parents'/carers' (hereafter referred to as "parents") views and experiences of managing sleep problems in children with ADHD 2) barriers and facilitators to engagement with target behavioural approaches to inform the development of a digital sleep guide.

Research questions: what are parents':

1. Views and experiences of behavioural sleep management strategies?
2. Experiences of seeking information about managing sleep problems?
3. Perceptions about a potential digital sleep guide for managing sleep problems in children with ADHD?

We also explored the impact of having a child with ADHD with sleep problems; themes related to this are reported separately [19].

2. Material and methods

This study is reported using the Standards for Reporting Qualitative Research [25].

2.1. Design

A qualitative semi-structured interview study.

2.2. Epistemological stance, research team and reflexivity

The research team included academics and clinicians with expertise in paediatric sleep medicine, child and adolescent psychiatry, health psychology, qualitative research methods and primary care research.

Reflexive notes and diary entries were made throughout the study and data collection/analysis were discussed and developed iteratively within the multidisciplinary team to interpret the data from multiple perspectives. The data was approached with a pragmatic stance, to describe parents' views/experiences of the techniques and practically inform the future development of an intervention. Personal experiences of being a parent, from both the clinical/academic team members and public contributors, may also have influenced interpretation.

2.3. Sample and recruitment

2.3.1. Eligibility criteria

Inclusion criteria were UK parents/carers of children aged 6-12 with both:

- Diagnosed ADHD or on a waiting list for an ADHD assessment.
- Chronic insomnia symptoms (CIS), i.e., difficulty falling asleep, maintaining sleep or early morning waking, for 3 or more nights a week, for at least 3 months.

Parents/carers were excluded if they were unable to understand/communicate in English or if the child had (1) severe learning difficulty, defined by whether their child had been recommended to attend a school for children with severe learning disabilities, (2) co-morbid mental or physical ill health or disability likely to require specialist supervision or frequent admission to hospital, or (3) other health problems which impacted sleep.

2.3.2. Sample size

A priori, it was anticipated that due to specific topic aims and a team consisting of experienced qualitative researchers, 15-25 interviews would provide sufficient information power for interpretation of the data [26]. This was reviewed as data collection and analysis progressed. Furthermore, after 17 interviews with 18 parents, the researchers judged the study had sufficient information power and data collection ended.

2.3.3. Recruitment and sampling strategy

The study recruited participants via social media (public posts and targeted advertising for parents with diverse characteristics), a local fostering organisation, word of mouth and snowballing sampling, with adverts directing interested individuals to an online form.

The online form checked participant eligibility and asked optional demographics questions to enable the research team to purposively select a maximally diverse sample. These included parent/carer characteristics (gender, ethnic background, highest level of qualification, relationship to child), child characteristics (age, gender, severity and type of insomnia) and household composition.

2.4. Data collection

Semi-structured qualitative interviews were conducted between March-August 2023 via telephone/videocall and lasted between 30 and 60 minutes. Fourteen interviews were conducted by SH (female post-doctoral Research Fellow and Psychologist, experienced in qualitative and child sleep research) and three by a master's student, supervised by KG/SH. Neither interviewer knew the participants and both presented themselves as non-clinical and neutral to the participants.

A semi-structured topic guide (supplementary information 1), developed based on academic and clinical research team expertise, guided the interview. Open-ended questions asked about 1) experiences of having a child with ADHD and sleep problems, 2) experiences of managing sleep problems, 3) views/experiences about individual behavioural sleep strategies as mentioned below, 4) experiences of seeking information and management support and 5) views about seeking information online and the potential development of a digital sleep guide.

Types of behavioural sleep strategies we individually asked about were:

- Consistent bedtime routines.
- Changing the sleep environment
- Encouraging positive sleep habits, including reducing screen us and caffeine, changing the timing of evening meals and increasing physical activity.
- Calming activities
- Bedtime fading
- Extinction-based sleep strategies
- Rewards for positive sleep behaviours

After the interview, further optional demographics questions asked about parent/carer characteristics (age, sexual orientation, employment status), child characteristics (ethnic background, diagnosed with ADHD or awaiting an assessment, diagnosed with ASD and whether they room share with another household member) and family composition (number of children in the household, number of others with ADHD in the household). Participants were given a £20 shopping voucher to thank them for their time.

Interviews were audio/video recorded and professionally transcribed, with transcripts checked by the research team for accuracy.

2.5. Data analysis

Data were handled in NVivo software and analysed iteratively based on a thematic analysis using the six steps outlined by Braun and Clarke [27]. This method was chosen due to its descriptive and theoretically flexible nature, to pragmatically inform future development of a digital sleep intervention. The data in this study were analysed on a semantic level, and this analysis was conducted inductively whereby codes and themes were developed from the data collected.

The researchers familiarised themselves with the data and SH made summary notes immediately following interviews. AR coded the transcripts, developed an initial coding framework and themes which were discussed with the wider team (SH, KG, IM, CMH, SC). AR's themes were developed initially for the aims regarding exploring the impact of sleep problems, described in Rowsell et al. [19]. In the context of the current paper, these were developed further by SH to align with the specific research aims of the current paper (regarding 1) parents' views and experiences of managing sleep problems in children with ADHD and 2) barriers and facilitators to engagement with target behavioural approaches to inform the development of a digital sleep guide). For this paper, the themes were developed with the inductive and descriptive approach to thematic analysis described above. They were then revised and finalised with KG using summaries/coding frameworks and subsequently the wider team (AR, IM, CMH, SC).

Throughout the analysis, data regarding individual sleep strategies, as mentioned in 2.4, fed into the main analysis, whereby common themes across the entire data set (including strategies) are given, and examples of specific strategies are discussed within the themes. To inform content specific to individual strategies in future interventions, more detailed and applied summaries of findings regarding individual strategies are given in the supplementary material.

2.6. Public and patient involvement and engagement

Public contributors (parents of children with ADHD) were involved in the design of study materials, ensuring they were relevant and accessible to participants. The topic guide and eligibility check questions were piloted with 2 public contributors prior to data collection, confirming timing, relevance and readability.

2.7. Ethical considerations

This study was approved by the University of Southampton Faculty of Medicine ethics committee (ERGO 78034). Participants provided informed consent prior to interview. Transcripts were pseudo-anonymised with participant identification numbers.

3. Results

3.1. Sample characteristics

One hundred and eighty-three parents expressed an interest in this study and 107 (58%) were eligible to take part. 18 parents took part in 17 interviews. Parent age ranged from 24 to 48 years (mean 39.1). The children they discussed were aged 6-12 years (mean 8.9). See Table 1/2.

3.2. Themes

Two descriptive themes, each with 3 subthemes (Fig. 1), common across techniques, highlighted key influences on engagement with sleep techniques and the support needs of parents.

Table 1
Parent/carer and household characteristics (N = 18).

		N	%
Gender	Female	16	88.9
	Male	2	11.1
Ethnicity	White British	14	77.8
	Asian or Asian British	2	11.1
	Mixed White & Black Caribbean	2	11.1
Sexual orientation	Heterosexual	14	77.8
	Lesbian or gay	2	11.1
	Bisexual	1	5.6
Relationship to child	Mother	16	88.9
	Father	1	5.6
	Step-father	1	5.6
Highest qualification	'A' level/equivalent (aged ≈ 18y)	6	33.3
	GSCE/O level (aged ≈ 16y)	2	11.1
	University Degree	4	24.2
	Postgraduate qualification	2	11.1
	Other/missing	4	24.2
Employment status	Full time	11	61.1
	Part time	4	22.2
	Other	3	16.8
Number of household parent/ carers	2 parent/carer household	12	66.7
	Single parent/carer household	6	33.3

Table 2
Child characteristics (N = 18).

		N	%
Gender	Female	7	38.9
	Male	11	61.1
Ethnicity	White British	10	55.6
	Asian or Asian British	2	11.2
	Mixed White & Black African	1	5.6
	Mixed White & Black Caribbean	1	5.6
	Any other mixed background	1	5.6
	Other	1	5.6
Parent reported sleep problems	Falling asleep or resisting bedtime	17	94.4
	Night-waking	14	77.8
	Early morning waking	12	66.7
Parent reported sleep problem severity	Moderate	11	61.1
	Severe	6	33.3
ADHD diagnosis	Diagnosed	12	66.7
	Awaiting assessment	6	33.3
Diagnosis of autism	Yes	2	11.1
	No	14	77.8
	No, but suspected	7	38.9
Room-shares with another household member?	Yes	6	33.3
	No	12	6.7

3.2.1. The challenges of engaging with sleep techniques

3.2.1.1. Parent/carer belief that behavioural sleep management techniques are ineffective for children with ADHD. Whilst parents were generally in favour of behavioural techniques, this theme describes an overarching parental scepticism about the effectiveness of behavioural techniques for children with ADHD. Participants often reported that they are challenging to implement and not always appropriate or effective for their child. Reasons included child resistance, hyperactivity at bedtime or a strong belief that the parent had tried everything with no success.

> "… where that [camping out] might work with a child without ADHD and they settle down, that's not how it works with a child with ADHD … the boundaries just aren't the same with them." (P22, mother of 9-year-old boy)

For example, whilst some parents liked the idea of bedtime fading, common concerns were that their child would not tire and go to sleep, regardless of bedtime delay.

> "… We've tried just letting him keep going until he crashes, but then we found we never got to that stage where we just wanted to crash, like, we would've just kept going all night long if the opportunity came … He wasn't able to recognise himself, that he was tired, and the more tired he became, you go into, he almost goes into that overdrive where we're overstimulated and, and just can't get ourselves out of that spiral" (P6, mother of 9-year-old boy)

Similarly, parents often stated that rewards for positive sleep behaviours would not work for their child for a variety of reasons: the child does not understand what is expected of them, their motivation is different to children without ADHD, or because they would not be able to wait for a reward. Smaller or more immediate rewards or rewards were suggested as something that could help.

> "that doesn't work for us, and I would say that - I would expect for a lot of ADHD kids it would be the, the same issue, because that route of motivation just isn't the same for us" (P14, mother of 8-year-old boy)

> "… I think perhaps I, I should have made them smaller goals almost …" (P25, mother of 11-year-old girl)

Most parents commented that increasing physical activity might worsen children's hyperactivity, rather than facilitate sleep.

> "I've done increasing physical activity before … it has not worked at all because he, he doesn't get worn out" (P8, mother of 11-year-old boy).

To help implementation of bedtime routines, participants often suggested that wind-down activities need to be short, motivating and calming. For example, stories or lego, rather than something that takes longer that their child may easily become bored with.

> "… it's reading a book for us … Erm, the other thing that she finds quite calming is story tapes … when she goes to sleep, just play a story tape and she, she enjoys that and I think that's quite calming. She, she wouldn't entertain yoga or anything because she can't stay focused." (P25, mother of 11-year-old girl)

Participants also often reported that current sleep advice is too vague for children with ADHD. They wanted meaningful advice tailored to their needs without assumption of their situation. Instead, they perceived health professional advice, particularly around bedtime routines, to be "like being told to suck eggs".

3.2.1.2. Challenges of implementing sleep strategies in everyday life for children with ADHD. All participants stated that behavioural strategies, whilst helpful in principle, are difficult to implement in practice. This theme discusses how they described multiple every-day barriers, not directly relating to ADHD, but which make it more difficult for children with ADHD. These include family composition (single versus two-parent households, differing parental roles e.g., more demands in single parent households), sibling room sharing, bedroom/household spaces, working patterns, holidays, visitors, financial constraints and their child having other neurodevelopmental conditions.

For example, whilst most acknowledged that screen time in the bedroom can be unhelpful to sleep (supplementary information 2), in some situations they were beneficial and comforting, particularly if the child had ASD or 'neurodivergent traits'. Some voiced that electronics can be calming/regulating, despite expressing concern that they might be judged by others for this opinion.

> "like my children have never had the TV or anything in their room, and all but one have never gone to bed with phones. My daughter who's a year older than [name], she goes to bed with her phone

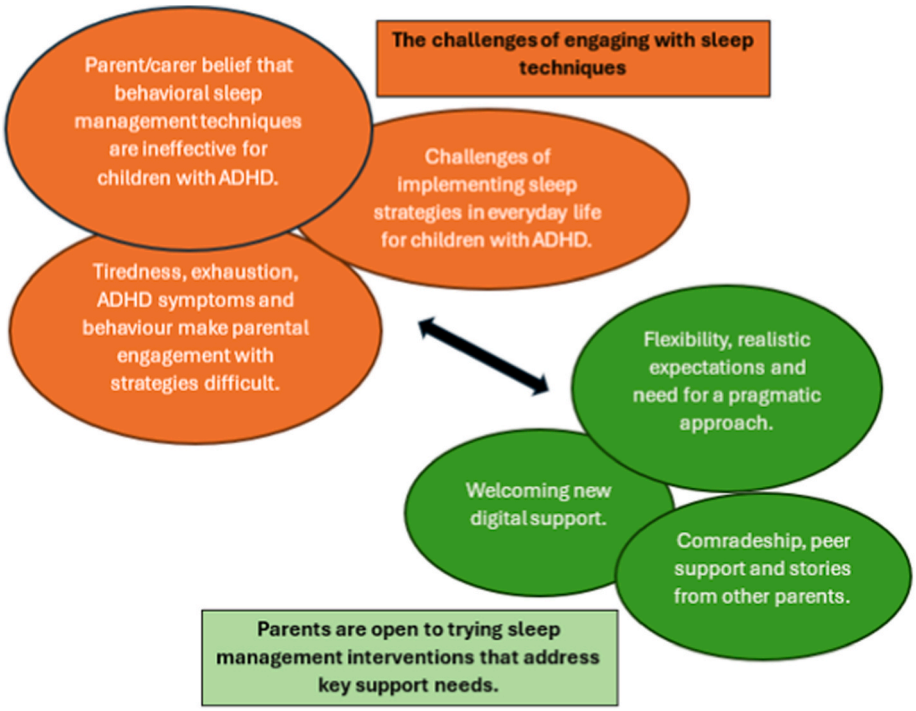


Fig. 1. Themes highlighting influences on engagement with sleep techniques and support needs.

'cause it's the only thing, like [name] does reading, it's the only thing that will settle her. And she has a lot of trouble sleeping as well. So I think, the majority of people, I think, electronics are bad, but sometimes that, it actually works in the opposite way for a very small minority … that's the only thing that I k-, that really kind of calms - she's a, a lot more kind of complicated. … lots of the advice is kind of, demonises the use of technology … And I do agree in most cases it probably doesn't help, but I think there is some people that it might help with … it's good advice to try not to have them, but in some cases, that might be the only thing that works, you know." (P2, mother of 11-year-old girl)

Similarly, there were mixed responses about gradually increasing physical distance (camping-out). Some agreed with the principles of this technique, however extinction-based strategies were generally perceived as difficult to implement or inappropriate due to their child's age, co-morbidities or individual/family circumstance (e.g., having an adopted child) and beliefs.

"we've tried a million different ways of doing it [extinction-based techniques] and it just doesn't help her … it doesn't work with the attachment … you have to work with, see what else the family's situation is" (P13, mother 10-year-old girl).

Bedtime fading was sometimes perceived as difficult to implement, due to worry about practical implications such as impact on parent evening routines (disrupting any small amount of time they have for themselves) or on sibling's evening routines. A few participants had not considered this method before but were interested to learn more and with the right information, potentially use it.

"… because I've moved homes … the three of them sleep in one room, it's not possible, 'cause then he's playing downstairs and the other two are, 'Why, why's he downstairs? Why's he downstairs?' Or, they'll deliberately egg each on by making a noise." (P12, mother of 9-year-old boy)

3.2.1.3. Tiredness, exhaustion, ADHD symptoms and behaviour make parental engagement with strategies difficult. This theme describes that a key barrier to parents' engagement with recommended sleep techniques was parental tiredness/exhaustion and their child's ADHD symptoms/behaviour impact on this. Alongside feeling they had tried everything with no success, these often seemed to lead parents to 'give in'.

"… you just get to a point of apathy, to be honest, where you just, like, do you know what, I just can't. I just haven't got the energy to, to try and be creative about how we're gonna do these things … You've got other - you've got your other kid to look after as well. There's only, there's only so much that you can try and follow, like, routines and strategies and whatever" (P15, father of 10-year old girl)

For example, participants viewed consistent bedtime routines positively but often acknowledged difficulty implementing them consistently without risking further tiredness and exhaustion. Things that had helped were perseverance, visual timetables, giving the child agency, peer support and setting realistic expectations.

"it's sticking to similar routines. I mean, like I say, I - in the holidays, it's not even just for them, it's for me. I have to allow them just to have telly at nighttime to chill sometimes, because I need that bit of respite as well" (P4, mother of 8-year-old boy)

"I think just all being on the same page … everyone kind of knowing what the routine is. And also, with the child, making them know it … for younger children, visual things are good … things to get ready in the morning, for instance, and it's, you know, like charts and things of, of what happens when. I know things like that work well". (P2, mother of 11-year-old girl)

Parents often reported a sense of hopelessness around implementing sleep techniques and reported the need for mental and physical health considerations for both parent and child. Some parents also reported their own ADHD diagnosis/symptoms made the situation more challenging or made it difficult to engage with techniques.

"… I have ADD, so I really struggle at home, erm, with my sleep anyway. So, I am like that person that cannot get out of bed. Whereas he can jump out of bed and he's like, 'Right, let's go there.' … obviously if I'm tired it makes me quite ratty, erm, and then obviously I've

got a toddler … so, you're never stopping … So, by the evening when I'm sat on the couch, like, shrivelled, shrivelled mess because I'm so tired" (P1, mother of 8-year-old boy)

3.2.2. Parents are open to trying sleep management interventions that address key support needs

3.2.2.1. Flexibility, realistic expectations and need for a pragmatic approach. All participants emphasised the need for pragmatic approaches to sleep techniques incorporating flexibility in choosing/implementing them and realistic expectations (particularly for strategies they were sceptical about). Although parents were positive about, and used some techniques (e.g., bedtime routines, calming activities, changing the sleep environment, reducing screen time), a pragmatic approach was emphasised as key.

"one thing that would be useful would just be, like, more strategies, you know, and different things that you can use, and maybe how you can mix and match …" (P15, father of 10-year-old girl)

"if something doesn't work, try and adapt it …" (P4, mother of 8-year-old boy)

For example, parents often described that consistent bedtime routines may not always be possible (e.g., parent/carer working patterns, special occasions or when children live between two households where parents live separately).

"… in an ideal world, we'd all love to stick to a routine to, to the T, wouldn't we, but the thing is, is life happens all the time … now I'm starting a full-time job, so unless I wait, obviously, till I got in a routine, I know roughly what times we'll be back and what times we'll be doing things … I could, kind of, set up some sort of similar routine. I think the only downfall with that is, like I say, is if they expect something at a certain time and you're not doing it, that could then cause anxiety … say if you've got to go out somewhere, you know, and it's an accident, or you've got something particular on … You can't do your routine because that's gonna change it, and then that's gonna make them more anxious. And then they're gonna not want to sleep …" (P4, mother of 8-year-old boy)

3.2.2.2. Welcoming new digital support. Most participants had searched for information online; including from NHS resources, ADHD websites and social media. However, they often had mixed opinions about them, suggesting they were inaccessible, untrustworthy, or not relevant to their child with ADHD.

"… you just want to see what's available out there, what, what support you can get. Like anything, there's always a massive waiting list." (P13, mother of 10-year-old girl)

The parents therefore welcomed development of digital sleep support if it accommodated key needs. Important elements to include were holistic and relevant information about ADHD and sleep problems, normal sleep patterns, use of medication (including side effects), tips for children, reassurance that a child with ADHD can have a 'normal life', comorbid conditions (for example autism spectrum disorder), accessible and engaging language and signposting. Off-putting features described included use of jargon, content perceived as inauthentic, prescriptive or inflexible.

"… if I had more understanding of how ADHD actually affects them, and how, obviously, it can affect their sleep and their appetite, I think if I had more of that information, I'd have more understanding of why they do it." (P27, mother of 7-year-old boy)

"if I've looked at something and then I can find other research then th-, that's, that would make me feel that it's more reliable." (P22, mother of 9-year-old boy)

They suggested that a credible, friendly and trusted digital support guide could fill the online information and health professional access void. Their needs were wider than simply practical advice. They wanted reassurance; that that they are not alone, that sleep can get better and to realistically say 'try it, if it doesn't work it might not be right for your child'.

3.2.2.3. Comradeship, peer support and stories from other parents. All parents highlighted peer support needs, welcoming experiences and support from other parents of children with ADHD. Many had sought support from family, friends and peers, with mixed experiences, but often described the importance of drawing on experiences of others in similar situations.

"Parents always like to hear how other parents get on whether they're good or bad, you know. They don't always want to hear shining stories of, 'Oh, and yes and my daughter slept beautifully after having a weighted blanket on her back for two nights,' you know, they want to hear both success and maybe not so good stories and how, how to overcome them. Or, how, what worked well and what didn't work well." (P13, mother of 10-year-old girl)

Participants described that stories from other parent of children with ADHD would provide examples of individual strategies, but importantly would validate their feelings, reinforce that they are not alone, and promote trust in strategies. Stories would provide hope, acknowledge parents' needs and encourage them to look after themselves. This was perceived as a contrast to the judgment parents/carers felt from the outside world, including professionals.

"some sort of support mechanism-, you know, or signposting to support for parents where they can realise that, you know, there's loads of other people going through similar stuff, and that actually, you're doing a really good job" (P15, father of 10-year-old girl)

"it's reassurance for the parents … this isn't your fault, you are not a failure at being a parent, because I'd be like, well, she just - why won't she go to sleep? I'm doing everything right, everybody's said that this should get her to sleep, and if somebody had just gone, 'It's not you,' that would have been really helpful" (P9, mother of 6-year-old girl)

3.3. Summary of thoughts about individual sleep techniques

To pragmatically inform the development of our digital intervention, we also explored parental experiences and views about individual behavioural sleep techniques. Further details for individual techniques and quotes are given in supplementary information 2.

4. Discussion

This study provided novel insights into the challenges parents of children with ADHD face when looking for and engaging with behavioural sleep techniques to improve their child's CIS, a topic that has been previously under-researched.

Like previous research [18], parents generally valued behavioural sleep techniques (particularly consistent bedtime routines, regulators/comforters and limiting things that are unhelpful to or hinder sleep). French et al. [18], similarly, found that some strategies were more valuable than others, albeit with mixed success, and that sometimes they are challenging to implement. These findings are also echoed in more recent qualitative research with 21 child participants with ADHD who highlighted the importance of routines and a calming environment [28]. Although parents in the current study placed less importance on physical

activity than the participants in French et al. [18] and Alvstrand et al. [28].

However, the current study also uniquely identified further challenges to engaging with sleep techniques; commonly the perception that they are ineffective for children with ADHD. Nonetheless, an umbrella review of meta-analyses of randomised trials exploring the efficacy of sleep interventions for children indicates that sleep techniques are effective for children with ADHD, albeit with small-to-moderate effect sizes and limited certainty of evidence [13]. Given that many parents in the current study also felt advice was not specific enough to children with ADHD, more parental support and the way information is given to parents may influence their motivation and perception of the effectiveness and appropriateness of these techniques. More reassurance and clarity could be provided to parents about the principles behind the recommended sleep management techniques and why they could help their child with ADHD.

For example, whilst the bedtime fading technique may not be appropriate for every child, there were common reservations related to parental worry about the impact of their child going to bed later and the belief that their child would remain active no matter the time of night. Information about how to implement this technique in children with ADHD, which children this would be appropriate for, and reassurance on the long-term benefits may support and empower parents/carers to implement the technique at home. There is evidence from other parenting programmes (sleep and non-sleep-specific) for children with ADHD that parent mediated behaviour change in the child can be achieved [15,16,23,29,30].

Another common barrier expressed by parents was a perception that techniques were inflexible and not easy to use in the real world. Parental considerations included 1) increased impulsivity and hyperactivity in their child (therefore resistance to routine, rewards, new behavioural strategies and an inability to switch-off); 2) individual child and family situation and 3) other needs for comorbid disorders, such as ASD. These further demonstrate the importance of tailoring interventions to the child and family's specific needs. An Australian randomised controlled trial tested a practitioner led intervention for managing behavioural sleep problems in children with ADHD; 'Sleeping Sound with ADHD'. The intervention included sleep education and flexible, but tailored, information and was effective (for reducing parent-reported ADHD symptoms) with samples of <200 children from pediatric practices [16] and <300 children in the community [15].

Other important considerations for engaging with techniques in the real world included parental exhaustion and mental health. As reported in Rowsell et al. [19] parental exhaustion often led to parents giving in to sleep problems. Based on our analysis, this led to giving up on management techniques and highlighted the need to encourage persistence and offer reassurance to parents. Furthermore, parents/carers in the current study often reported they themselves have ADHD or other neurodiversity. As ADHD is characterised by high heritability [31], parent-mediated interventions including these sleep techniques must also accommodate the needs of parents who themselves may have inattention and may struggle to self-regulate.

Parents/carers further highlighted support needs. In line with previous qualitative research of parents' experiences seeking help for young children (not specific to ADHD) [32,33], parents in the current study perceived much current health professional and online advice to be lacking, or to be impersonal or patronising. Parents in the current study often reported a void in information [19] which also often resulted in the use of online resources. As a consequence, they endorsed the development of an evidence-based, relevant and flexible digital resource for the management of their child's sleep problems.

4.1. Implications

4.1.1. Clinical practice and clinical resources

Generally, information around behavioural sleep strategies needs to be tailored to families of children with ADHD and diverse family situations. Similarly to information included in some available programmes such as the Sleeping Sound with ADHD intervention [15,16], clear and relevant but flexible information from both health professionals and support resources are needed to equip parents with the knowledge and confidence to implement and persist with appropriate strategies effectively.

Parents/carers may also benefit hearing from other parents who have been in similar situations. This will validate their feelings, reassure them that they are not alone and help build parental self-efficacy and trust in the strategies they are being advised to try and set realistic expectations. These needs are similarly reported in other studies highlighting the needs of parents of children (not specific to ADHD) [33].

4.1.2. Future research

Parents in this study were supportive of the development of a future digital sleep guide. It should address key challenges highlighted by parents in this study and include the key important elements for parents regarding tailored information and peer support. Notably, tailored information and peer support through parent quotes have been used in development of other parent/carer child health interventions such as the management of childhood eczema [34]. Digital intervention should also address challenges with effective and timely implementation of the programmes.

Future research with more fathers and parents from ethnic minority backgrounds would also be important to ensure the development of resources sensitive to the needs of diverse parents/carers.

Findings in this study were used to inform the development of the DISCA intervention 'Sleep Buddy' [35] whereby the intervention incorporates reassurance and support for parents, advice specific for children with ADHD and different family circumstances, and peer support.

4.2. Strengths and limitations

This study highlights important considerations for the implementation of behavioural sleep strategies for parents of children with ADHD. However, it must be noted that ADHD diagnosis was either self-reported by the parent or to be confirmed. Nonetheless recent UK data from a large child mental health service reported 94% of such children, when assessed, indeed receive an ADHD diagnosis [36]. Whilst it may be possible that not all children would receive a full ADHD diagnosis, this mimics the real-world challenges of families on long waiting-lists and the importance of providing support to these families who reported similar emotional turmoil in managing their child's sleep problems.

The families in this sample had a wide range of characteristics, including adoptive parents, same-sex parents, single versus two parent households, perceived sleep problem severity, age and child gender. However, it would have been relevant to include more fathers, families from ethnic minority backgrounds and parents/carers with more varied levels of education.

Given the focus of our study on behavioural interventions and the need to limit the number of sampling/demographic questions, we did not gather information on children's medication status.

5. Conclusion

Parents of children with ADHD are generally supportive of using behavioural sleep techniques for their child's CIS. However, they often perceive them as ineffective or challenging to implement. Parents require tailored, realistic but flexible and reassuring guidance. They welcome improved evidence-based digital support, while highlighting the importance of embedded validation and experiences from other parents in similar situations.

Disclosures

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CRediT authorship contribution statement

Samantha J. Hornsey: Data curation, Formal analysis, Investigation, Methodology, Project administration, Writing – original draft. **Alison Rowsell:** Data curation, Formal analysis, Methodology, Software, Writing – review & editing. **Catherine M. Hill:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – review & editing. **Samuele Cortese:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – review & editing. **Kate Greenwell:** Methodology, Supervision, Writing – review & editing. **Ingrid Muller:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Catherine Hill reports a relationship with AGB Pharma that includes: consulting or advisory. Catherine Hill reports a relationship with Neurim Pharmaceuticals Ltd that includes: consulting or advisory. Catherine Hill reports a relationship with National Institute for Health Research Programme Grants for Applied Research that includes: funding grants. Catherine Hill reports a relationship with The Chinese University of Hong Kong that includes: speaking and lecture fees and travel reimbursement.Samuele Cortese reports financial support was provided by National Institute for Health and Care Research. Samuele Cortese reports a relationship with National Institute for Health and Care Research that includes: funding grants. Samuele Cortese, NIHR Research Professor (NIHR303122) is funded by the NIHR for this research project. The views expressed in this publication are those of the author(s) and not necessarily those of the NIHR, NHS or the UK Department of Health and Social Care. Samuele Cortese is also supported by NIHR grants NIHR203684, NIHR203035, NIHR130077, NIHR128472, RP-PG-0618-20003 and by grant 101095568-HORIZONHLTH- 2022-DISEASE-07-03 from the European Research Executive Agency. Prof. Cortese has declared reimbursement for travel and accommodation expenses from the Association for Child and Adolescent Central Health (ACAMH) in relation to lectures delivered for ACAMH, the Canadian AADHD Alliance Resource, the British Association of Psychopharmacology, Healthcare Convention and CCM Group team for educational activity on ADHD, and has received honoraria from Medice.If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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